



Post-Quantum Cryptography Migration

Part 18 - Finding & Governing It: Developer & Migration Tooling

Compiled by the Spaced Research Ledger

Last updated: 2026-08-29

Before an organization can migrate anything, it has to find its cryptography, and almost none can. Asking which systems use RSA is a question most enterprises cannot answer in minutes, or at all. The tools that fix that, inventories, scanners, validators, and test rigs, are this report's subject.

Five sections cover the toolchain. Section 1 covers the cryptographic bill of materials, Section 2 the commercial discovery scanners, and Section 3 the government validation programs. Section 4 covers interoperability and test tooling, and Section 5 performance and side-channel research.

The format war is over and the mandate arrived. The cryptographic bill of materials, CBOM, is an internationally ratified standard inside CycloneDX. The June 2026 executive order made it a federally defined artifact, with minimum elements due from CISA in 2027. The scanner market consolidated around it, with a billion-dollar funding round and IBM scanning 47 million of its own lines of code as the proof case.

The measurement layer carries this collection's sharpest warning. One paper recovered a full ML-KEM secret key from fifteen power traces in under ten minutes. The algorithms are sound, and unprotected implementations leak anyway, which is why the tooling in Section 5 exists.

1. Crypto Discovery & CBOM

A cryptographic bill of materials is an inventory of every algorithm, key, and certificate a piece of software uses, machine-readable and exchangeable like the software bills of materials before it. In two years it went from a proposal to a ratified standard to an executive-order requirement. This section covers that ascent.

Recent Developments

The executive order made CBOM policy. Executive Order 14412, signed June 22, 2026, is the first federal directive naming CBOM as a defined artifact, ordering minimum required elements within 270 days [1]. It requires agencies to migrate high-value assets to post-quantum key establishment by the end of 2030 [1]. The standard beneath it was already ratified: CycloneDX 1.6, carrying native CBOM support, published as an ECMA standard in 2024 [1].

Version 1.7 added provenance and a wider cryptography registry [2], and version 2.0 is announced for fall with threat models and controls as exchangeable data [3]. NIST's discovery practice guide remains in



preliminary draft [4]. The tool ecosystem keeps pace, with commercial platforms adding readiness classification [5] and multiple tools generating classified CBOMs across dozens of algorithms [6]. CISA's final SBOM minimum elements require post-build artifact scanning and narrow the accepted formats [7][7].

Current State

The lineage is short and open-source. CBOM began as an OWASP working-group proposal in October 2023, folding IBM's open-sourced project into CycloneDX [8]. It models cryptographic assets as a first-class component type with dependency relationships, producible by existing build tooling [9]. IBM's research paper places it in a wider bill-of-materials ecosystem aimed at full-stack transparency [10]. The reference documentation is public [11], and an IETF draft defines an inventory output schema built on it [12].

The practical advice is consistent: discovery is the slowest part, mature SBOM programs have a head start, and starting before the federal guidance lands is the right call [1]. Open-source kit generates a basic CBOM fastest, with commercial platforms layering break-year forecasts and compliance mapping [13], all in service of finally answering the RSA question [13].

2. Commercial Discovery Scanners

A market of scanners now hunts cryptography through source code, binaries, networks, and cloud workloads. Its short history is a consolidation story: the pioneering startups were acquired, the acquirers raised big, and the tools are converging on CBOM as their output format. This section covers the three main camps.

Recent Developments

The market matured visibly this summer. IBM's Quantum Safe Explorer shipped successive releases adding Python analysis, JavaScript compliance reporting, and improved detection through loops and arrays [14][15], with expanded API discovery and customizable severities [16]. A security bulletin patched vulnerabilities in the scanner's own dependencies across earlier versions [17]. IBM ate its own cooking, scanning 5,815 repositories and 47 million lines of code, surfacing thousands of findings and generating CBOMs [18]. It is standing up a quantum-safe center in Singapore around the tooling [19].

Keyfactor shipped AgileSec 3.6 with post-quantum detection and CBOM export improvements [20]. It raised more than a billion dollars to accelerate the platform [21], and extended its partner program with lab access [22]. SandboxAQ aligned itself publicly with the executive order [23].

Current State

The camps trace back to two acquisitions. Cryptosense, the Inria spin-off whose analyzer covered both static code and runtime invocation, reportedly sat behind most EU interbank messaging compliance before SandboxAQ bought it [24]. SandboxAQ fused it into an integrated suite spanning filesystem, application, and network inventory [25], and joined NIST's migration project as an official collaborator [24]. It published a public database of cryptographic risk across open-source software with nearly a billion entries [26]. The product has since expanded toward AI and non-human identity posture [27].

Keyfactor built its position by buying discovery. It acquired InfoSec Global and CipherInsights in 2025, folding cryptographic discovery and no-code-change agility management into its platform [28][29], and added workflow integration through ServiceNow [30]. It partnered with IBM Consulting on end-to-end discovery, CBOM construction, and prioritization [31]. It also keeps a free sandbox for post-quantum certificate testing alongside its stewardship of Bouncy Castle [32]. IBM's own scanner rounds out the field, extended to mainframe Linux [33] and feeding its cryptography-management product through import profiles [34].

3. Algorithm Validation Programs

Before a cryptographic implementation can be sold to the U.S. government, its algorithms pass automated validation. The testing machinery for the post-quantum algorithms is now routine infrastructure, which is its own kind of milestone. This section is short because the system simply works.

Recent Developments

The specifications keep pace with the standards. The automated validation protocol's ML-DSA test specification published in August, covering key generation, signing, and verification under the standard and its technical revision [35]. The validation search already lists ML-KEM and ML-DSA validations through June 2026, including silicon implementations [36].

Current State

The machinery is complete end to end. The ML-KEM test specification defines its schema [37], and the validation server supports the technical-revision test modes including both private-key formats [38]. The prerequisite tables wire the algorithms into the broader validation graph [39], and full test vectors are public [40].

4. Interop & Test Tooling

Standards only matter if implementations actually talk to each other, and a small ecosystem of test servers, scanners, and dissectors exists to prove it. Much of it traces to one university project that became the field's shared infrastructure. This section covers how post-quantum interoperability actually gets checked.

Recent Developments

The scanning tools went mainstream. pqcscan, an open-source scanner, checks TLS and SSH servers for post-quantum algorithm support with exportable reports [41], and is referenced in NIST's own migration FAQ [42]. A free web tool performs a real hybrid handshake against any public server [43]. Wireshark fixed its reporting of post-quantum signature algorithms [44], and OpenShift's TLS scanner gained a dedicated post-quantum readiness check with a failing exit code [45].

Current State

Open Quantum Safe is the shared test bench. Its public interoperability server exposes a port per algorithm combination, explicitly for test-driving rather than production [46]. Its C library ships benchmarking as a first-class feature [47]. The institutional base is broad, founded at Waterloo with contributors and funders across the major vendors, now under the Linux Foundation's post-quantum alliance [48].

Its demo pipeline retests the third-party integrations weekly [49], with one documented gap where the BoringSSL fork's algorithm set has drifted from the rest [49]. The IETF Hackathon's certificate project complements it, a compatibility-matrix repository where vendors upload artifacts validated by automation, feeding results to standards groups [50]. NIST frames the whole ecosystem as how interoperable post-quantum reaches commercial products [51]. Even the venerable testssl.sh has an open request to call out missing post-quantum support [52].

5. Performance & Side-Channel Tooling

The last mile of trust is measurement: how fast the algorithms really run, and whether their implementations leak secrets through power draw and timing. The benchmark numbers are increasingly encouraging. The side-channel results are the collection's starkest warning. This section covers both.

Recent Developments

The measurement infrastructure keeps current. The SUPERCOP benchmark suite carries current ML-KEM results with constant-time checking flags [53]. A defense-research framework proposes standardized side-channel testing for ML-KEM in military communications, warning the algorithms' benefits are wasted without countermeasure evaluation [54]. A unified masked hardware implementation of ML-DSA and ML-KEM published with formal proofs and no detected first-order leakage [55], while liboqs retired a legacy constant-time test in its cleanup [56].

Current State

The performance news is mostly good. On the weakest common microcontroller, ML-KEM-512 completes a key exchange in 36 milliseconds at a fraction of classical ECDH's energy [57]. ML-DSA signing shows wild latency variance from rejection sampling, with worst percentiles over a second [57]. At the opposite scale, GPUs push millions of key generations per second for telecom loads [58]. A 5G handset study finds ML-DSA cheapest on CPU and latency while the hash-based combinations miss the strictest latency budgets [59].

AWS's first-party numbers put hybrid overhead at kilobytes and microseconds, amortizing toward zero with connection reuse [60]. The constrained-platform caveat stands: classical X25519 still wins on a Raspberry Pi, reflecting immature ARM optimization, with conservative alternatives paying more [61]. Pre-silicon verification methods now certify masked hardware structurally [62].

The side-channel results demand attention. Full ML-KEM key recovery from fifteen power traces in under ten minutes defines the gap between algorithmic and implementation security [63]. The same work proposes adaptive defense, detecting attacks in real time and scaling countermeasures at a fraction of always-on

masking's cost [63]. A CRYPTO paper showed masking alone does not guarantee safety, attacking an intermediate value in masked ML-DSA on physical devices [64].

A template attack builds its leakage profiles from public data on the target itself, no clone device required [65]. The community response is infrastructure. Public trace datasets support reproducible leakage research [66], and one study frames ciphertext malleability as both attack surface and countermeasure [67]. Mentorship work is wiring constant-time analysis into continuous integration [68], and formal verification of vectorized sampling is headed for Signal's backend [69]. Differential fuzzing found real bugs in the backup KEM [70], and a framework ran over a hundred million generated test cases across the major libraries [71].

About This Report

The Spaced Research Ledger is an automated system that gathers claims from live public sources, checks each one against its origin, and records every disagreement instead of merging it. Every stage runs on a different AI model, drawn from four to seven systems across competing labs, so no single model both finds a claim and judges it. The Spaced Research Ledger is designed and maintained by Ridley DiSiena.

Sources

1. cocode.com — cryptographic bill of materials cbom - <https://cocode.com/blog/cryptographic-bill-of-materials-cbom/> - as of 2026-07-07
2. CycloneDX vs SPDX: Which SBOM Format to Choose in ... - <https://www.interlynk.io/resources/cyclonedx-vs-spdx-sbom-format> - as of 2026-06-23
3. CycloneDX Newsroom - <https://cyclonedx.org/news/> - as of 2026-08-05
4. Cryptography Bill of Materials (CBOM) - <https://cyclonedx.org/capabilities/cbom/> - as of 2026-08-08
5. Announcing sbomify v26.7.0: The One That Gets Quantum-Ready | Sbmify - <https://sbomify.com/2026/07/07/announcing-sbmify-v26-7-0-the-one-that-gets-quantum-ready/> - as of 2026-07-07
6. CycloneDX Tool Center - <https://cyclonedx.org/tool-center/> - as of 2026-08-13
7. CISA's 2026 SBOM Minimum Elements Now Require Post-Build Data - OPSWAT - <https://www.opswat.com/blog/cisas-2026-sbom-minimum-elements-now-require-post-build-data> - as of 2026-07-29
8. owasp.org — CycloneDX Cryptography CBOM - <https://owasp.org/blog/2023/10/03/CycloneDX-Cryptography-CBOM> - as of 2023-10-03
9. github.com — CBOM - <https://github.com/IBM/CBOM> - as of 2024-04-09
10. research.ibm.com — the anatomy of cryptography bills of materials standardization and practice in cyclonedx - <https://research.ibm.com/publications/the-anatomy-of-cryptography-bills-of-materials-standardization-and-practice-in-cyclonedx> - as of 2026-05-10
11. en.wikipedia.org — Cryptographic bill of materials - https://en.wikipedia.org/wiki/Cryptographic_bill_of_materials
12. draft-dev-xipher-cbom-extension-00 - <https://datatracker.ietf.org/doc/draft-dev-xipher-cbom-extension/00/> - as of 2026-04-20

13. o3.security — cbom cryptographic bill of materials - <https://o3.security/academy/cbom-cryptographic-bill-of-materials> - as of 2026-06-07
14. What's new - IBM Documentation - <https://www.ibm.com/docs/en/quantum-safe/quantum-safe-explorer/2.x?topic=whats-new> - as of 2026-07-24
15. What's new - IBM Quantum Safe Explorer - <https://www.ibm.com/docs/en/quantum-safe/quantum-safe-explorer/2.x?topic=whats-new> - as of 2026-07-24
16. Empowering CIOs to accelerate crypto-agility with IBM Quantum Safe Explorer - <https://www.ibm.com/new/product-blog/empowering-cios-to-accelerate-crypto-agility-with-ibm-quantum-safe-explorer> - as of 2026-08-15
17. Security Bulletin: IBM Quantum Safe Explorer is affected by multiple vulnerabilities - <https://www.ibm.com/support/pages/security-bulletin-ibm-quantum-safe-explorer-affected-multiple-vulnerabilities> - as of 2026-07-26
18. Empowering CIOs to accelerate crypto-agility with IBM Quantum Safe Explorer - <https://www.ibm.com/jp-ja/new/product-blog/empowering-cios-to-accelerate-crypto-agility-with-ibm-quantum-safe-explorer> - as of 2026-08-15
19. Singapore Institute of Technology and IBM Aim to Establish Quantum-Safe Centre that Addresses Post-Quantum Risks and Advances Singapore's Quantum-Safe Resilience - <https://www.prnewswire.com/apac/news-releases/singapore-institute-of-technology-and-ibm-aim-to-establish-quantum-safe-centre-that-addresses-post-quantum-risks-and-advances-singapores-quantum-safe-resilience-302857243.html> - as of 2026-08-21
20. AgileSec 3.6 Gives Security Teams More Visibility, More Control, and a Faster Path to Cryptographic Agility - <https://www.keyfactor.com/blog/agilesec-3-6-gives-security-teams-more-visibility-more-control-and-a-faster-path-to-cryptographic-agility/> - as of 2026-07-06
21. Keyfactor Scores \$1 Billion+ Investment for AI, Post-Quantum Security - <https://www.securityweek.com/keyfactor-scores-1-billion-investment-for-ai-post-quantum-security/> - as of 2026-07-07
22. Keyfactor Expands Global Partner Program to Equip Partners for AI and Post-Quantum Cryptography Transition - <https://thequbitreport.com/security-policy/2026/07/22/keyfactor-expands-global-partner-program-to-equip-partners-for-ai-and-post-quantum-cryptography-transition/> - as of 2026-07-22
23. SandboxAQ Supports White House Executive Order as Post-Quantum Cryptography Moves to Implementation - <https://www.prnewswire.com/news-releases/sandboxaq-supports-white-house-executive-order-as-post-quantum-cryptography-moves-to-implementation-for-national-security-302807077.html> - as of 2026-06-22
24. eetimes.com — sandboxaq buys cryptosense to deploy post quantum cryptography - <https://www.eetimes.com/sandboxaq-buys-cryptosense-to-deploy-post-quantum-cryptography/> - as of 2023-08-14
25. sandboxaq.com — sandboxaq unveils end to end security suite delivering advanced cryptographic management to the global 1000 and governments - <https://www.sandboxaq.com/press/sandboxaq-unveils-end-to-end-security-suite-delivering-advanced-cryptographic-management-to-the-global-1000-and-governments> - as of 2023-04-19

26. SandboxAQ Unveils OpenCryptography.com: The First Public Database Exposing Cryptographic Risks in Open-Source Software - <https://www.sandboxaq.com/press/sandboxaq-unveils-opencryptography-com-the-first-public-database-exposing-cryptographic-risks-in-open-source-software> - as of 2025-10-30
27. SandboxAQ AQtive Guard - <https://www.sandboxaq.com/solutions/aqtive-guard> - as of 2026-03-03
28. Keyfactor Acquires InfoSec Global and CipherInsights to Lead the Market in Quantum-Safe Security - <https://www.keyfactor.com/press-releases/keyfactor-acquires-infosec-global-and-cipherinsights/> - as of 2025-05-13
29. Keyfactor Infosec Global Cipherinsights Acquisition - <https://www.keyfactor.com/infosec-global-cipherinsights-acquisition/> - as of 2025-05-13
30. AgileSec and ServiceNow Enable Enterprise Quantum-Readiness with Cryptographic Posture Management - <https://www.keyfactor.com/blog/agilesec-and-servicenow-enable-enterprise-quantum-readiness-with-cryptographic-posture-management/> - as of 2025-10-30
31. keyfactor.com — keyfactor and ibm consulting launch joint solution to accelerate enterprise quantum safe transformation - <https://www.keyfactor.com/press-releases/keyfactor-and-ibm-consulting-launch-joint-solution-to-accelerate-enterprise-quantum-safe-transformation/> - as of 2026-01-28
32. businesswire.com — en - <https://www.businesswire.com/news/home/20240314254136/en>
33. Extending quantum-safe visibility to LinuxONE: IBM Quantum Safe™ Explorer now available with CLI support on Z-Linux - <https://community.ibm.com/community/user/blogs/vinaya-patil/2025/10/08/extending-quantum-safe-visibility-to-linuxone> - as of 2025-10-08
34. IBM Guardium Cryptography Manager announcement - <https://community.ibm.com/community/user/blogs/ivana-pham/2026/03/26/we-are-excited-to-announce-ibm-guardium-cryptograp> - as of 2026-03-27
35. ACVP ML-DSA JSON Specification - <https://pages.nist.gov/ACVP/draft-celi-acvp-ml-dsa.html> - as of 2026-08-14
36. Cryptographic Algorithm Validation Program CAVP - <https://csrc.nist.gov/projects/cryptographic-algorithm-validation-program/validation-search> - as of 2026-08-12
37. ACVP ML-KEM JSON Specification - <https://pages.nist.gov/ACVP/draft-celi-acvp-ml-kem.html>
38. Releases · usnistgov/ACVP-Server - <https://github.com/usnistgov/ACVP-Server/releases>
39. Cryptographic Algorithm Validation Program CAVP - Prerequisites - <https://csrc.nist.gov/projects/cryptographic-algorithm-validation-program/prerequisites>
40. PQC Archive - Post-Quantum Cryptography - <https://csrc.nist.gov/projects/post-quantum-cryptography/pqc-archive>
41. anvilsecure/pqcscan: Post-Quantum Cryptography Scanner - <https://github.com/anvilsecure/pqcscan> - as of 2026-08-14
42. Frequently Asked Questions about Post-Quantum Cryptography - <https://pages.nist.gov/nccoe-migration-post-quantum-cryptography/> - as of 2026-08-13
43. Post-Quantum Server Test | Myles Agnew - <https://www.mylesagnew.com/pqctest/> - as of 2026-08-25
44. Wireshark 4.6.7 Release Notes - <https://www.wireshark.org/docs/relnotes/wireshark-4.6.7.html> - as of 2026-08-12

45. tls-scanner module - github.com/openshift/tls-scanner - <https://pkg.go.dev/github.com/openshift/tls-scanner> - as of 2026-08-25
46. test.openquantumsafe.org - <https://test.openquantumsafe.org/>
47. [openquantumsafe.org](https://openquantumsafe.org/liboqs/) — liboqs - <https://openquantumsafe.org/liboqs/>
48. [github.com](https://github.com/open-quantum-safe/liboqs) — liboqs - <https://github.com/open-quantum-safe/liboqs>
49. [github.com](https://github.com/open-quantum-safe/oqs-demos) — oqs demos - <https://github.com/open-quantum-safe/oqs-demos>
50. [github.com](https://github.com/IETF-Hackathon/pqc-certificates) — pqc certificates - <https://github.com/IETF-Hackathon/pqc-certificates>
51. [nist.gov](https://www.nist.gov/pqc) — pqc - <https://www.nist.gov/pqc> - as of 2025-12-03
52. [Feature request] TestSSL pointing out missing support for PQC/KEM (NIST FIPS 203) - <https://github.com/testssl/testssl.sh/issues/2960> - as of 2025-12-08
53. Measurements of key-encapsulation mechanisms - <https://bench.cr.yp.to/results-kem.html> - as of 2026-08-05
54. papers.academic-conferences.org — 4907 - <https://papers.academic-conferences.org/index.php/eccws/article/view/4907> - as of 2026-06-15
55. [peerj.com](https://peerj.com/articles/cs-4003/) — cs 4003 - <https://peerj.com/articles/cs-4003/> - as of 2026-07-03
56. [liboqs/RELEASE.md](https://github.com/open-quantum-safe/liboqs/blob/main/RELEASE.md) - <https://github.com/open-quantum-safe/liboqs/blob/main/RELEASE.md> - as of 2026-07-09
57. [arxiv.org](https://arxiv.org/abs/2603.19340v1) — 2603.19340v1 - <https://arxiv.org/abs/2603.19340v1> - as of 2026-03-19
58. [arxiv.org](https://arxiv.org/pdf/2510.19982) — 2510.19982 - <https://arxiv.org/pdf/2510.19982>
59. [arxiv.org](https://arxiv.org/pdf/2507.17074) — 2507.17074 - <https://arxiv.org/pdf/2507.17074>
60. [factualminds.com](https://www.factualminds.com) — aws kms post quantum cryptography ml kem ml dsa - <https://www.factualminds.com/blog/aws-kms-post-quantum-cryptography-ml-kem-ml-dsa/>
61. [arxiv.org](https://arxiv.org/pdf/2605.06881) — 2605.06881 - <https://arxiv.org/pdf/2605.06881>
62. [arxiv.org](https://arxiv.org/pdf/2604.15249) — 2604.15249 - <https://arxiv.org/pdf/2604.15249>
63. [researchsquare.com](https://www.researchsquare.com/article/rs-9533349/v1) — v1 - <https://www.researchsquare.com/article/rs-9533349/v1> - as of 2026-05-13
64. [eprint.iacr.org](https://eprint.iacr.org/2025/276) — 276 - <https://eprint.iacr.org/2025/276> - as of 2025-06-04
65. [eprint.iacr.org](https://eprint.iacr.org/2026/056) — 056 - <https://eprint.iacr.org/2026/056> - as of 2026-01-16
66. [zenodo.org](https://zenodo.org/records/18681117) — 18681117 - <https://zenodo.org/records/18681117> - as of 2026-02-18
67. [arxiv.org](https://arxiv.org/pdf/2409.16107) — 2409.16107 - <https://arxiv.org/pdf/2409.16107>
68. In Open Quantum Safe LFX Mentorship – Enhancing Constant-Time Analysis Tooling in liboqs - <https://pqca.org/blog/2026/lfx-mentorship-enhancing-constant-time-analysis-tooling-in-liboqs/> - as of 2026-02-25
69. Formally verifying AVX2 rejection sampling for ML-KEM - <https://pqshield.com/post-quantum-cryptography-verification-for-ml-kem/> - as of 2025-12-17
70. Finding bugs in implementations of HQC, the fifth post-quantum standard - <https://blog.quarkslab.com/finding-bugs-in-implementations-of-hqc-the-fifth-post-quantum-standard.html> - as of 2025-03-21
71. A DIFFERENTIAL FUZZING FRAMEWORK FOR ML-KEM AND ML-DSA IMPLEMENTATIONS IN CRYPTOGRAPHIC LIBRARIES -

[https://oulurepo.oulu.fi/bitstream/handle/10024/63293/nbnfioulu-202606114145.pdf?](https://oulurepo.oulu.fi/bitstream/handle/10024/63293/nbnfioulu-202606114145.pdf?sequence=1&isAllowed=y)
sequence=1&isAllowed=y - as of 2026-05-01